



WATER APPLICATION VALUE ENGINE

WATER SOLUTIONS

WAVE Program Version: 1.85.12

Calculation Engine Version: 01.12.39.00

Database Version: 80



Project Name: SFC -CCRO-R85.4-M120 Exp SOAR 5000i_v1 r1
Case Name: TDS 85.4%
Customer: SFC
Prepared by: RN
Company: DuPont
Country: India
Date Created: January 08, 2026
Project Notes: Secondary treated STP--UF -CCRO

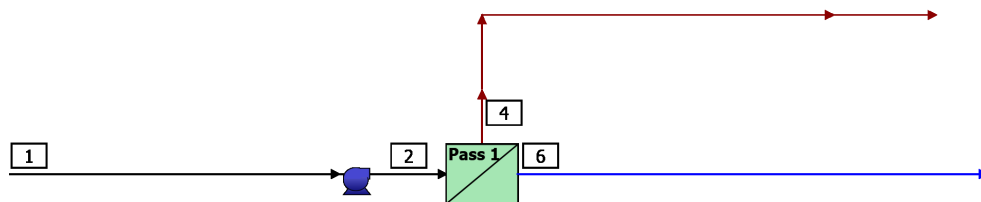
Case #: 1 of: 3
Case Notes: Case 1

Keywords:

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CCRO Detailed Report

RO System Flow Diagram



#	Description	Flow (m³/h)	TDS (mg/L)	Pressure (bar)
1	Raw Feed to RO System	636.8	601.0	0.0
2	Net Feed to Pass 1	636.8	601.0	12.2
4	Total Concentrate from Pass 1	92.9	4,028	10.2
6	Net Product from RO System	543.8	11.87	0.0

RO System Overview

Total # of Units	1	Online =	1	Standby =	0	RO Recovery	85.4 %
System Flow Rate	(m³/h)	Net Feed =	636.8	Net Product =	543.8		

Pass	Pass 1
Stream Name	Stream 1
Water Type	Waste Water (Conventional pretreatment,SDI<5)
Number of Elements	840
Total Active Area (m²)	31215
Feed Flow per Pass (m³/h)	636.8
Feed TDS ^a (mg/L)	601.0
Feed Pressure (bar)	10.8 - 12.2
Flow Factor Per Stage	0.70
Permeate Flow per Pass (m³/h)	543.8
Pass Average flux (LMH)	17.4
Permeate TDS ^a (mg/L)	11.87
Pass Recovery	85.4 %
Average NDP (bar)	8.7
Specific Energy (kWh/m³)	0.51
Temperature (°C)	20.0
pH	7.7
Chemical Dose	-
RO System Recovery	85.4 %
Net RO System Recovery	85.4%

Footnotes:

^aTotal Dissolved Solids and Conductivity includes ions, SiO₂ and B. It does not include NH₃ and CO₂

CCRO Overview

Pass		Pass 1
Elements per PV	Elements	7
Length Of PV	Elements	7
CC Recovery	(%)	42.36
PF Recovery	(%)	20.00
PF Feed Ratio	(%)	120.00
CC Concentrate Flow	(m³/h/pv)	7.00
PF Concentrate Flow	(m³/h/pv)	4.94
CC Net Feed Flow	(m³/h/pv)	12.15
PF Feed Flow	(m³/h/pv)	6.17
Total Cycles		8.62
PF Sequence Duration	(min)	1.90
CC Sequence Duration	(min)	10.24
Complete Cycle Duration	(min)	12.14
CC System Volume	(m³)	18.81

CCRO Sustainability

	Recovery	Concentrate			
		Volume		Value	
Traditional RO	00	5578368	m³/y	4630045	\$/y
CCRO	85	814417	m³/y	675966	\$/y
CCRO Savings		4763951	m³/y	3954079	\$/y

At water purchase price 0.14 \$/m³ and water disposal price 0.69 \$/m³

RO Flow Table (Stage Level) - Pass 1

Stage	Elements	#PV	#Els per PV	Feed				Concentrate			Permeate			
				Feed Flow	Recirc Flow	Feed Press	Boost Press	Conc Flow	Conc Press	Press Drop	Perm Flow	Avg Flux	Perm Press	Perm TDS
				(m³/h)	(m³/h)	(bar)	(bar)	(m³/h)	(bar)	(bar)	(m³/h)	(LMH)	(bar)	(mg/L)
PF	*FilmTec™ SOAR 5000iEE4	120	7	740.8	0.0	2.4	3.9	592.7	2.1	0.3	148.1	4.7	0.0	14.99
CC1	*FilmTec™ SOAR 5000iEE4	120	7	1,457	0.0	9.3	10.8	839.5	8.8	0.5	617.4	19.8	0.0	5.51
CC Final	*FilmTec™ SOAR 5000iEE4	120	7	1,457	0.0	10.7	12.2	839.4	10.2	0.5	617.5	19.8	0.0	18.01

RO Solute Concentrations - Pass 1

Concentrations (mg/L as ion)								
	Feed	Concentrate			Permeate			
		PF	CC1	CC Final	PF	CC1	CC Final	Total
NH ₄ ⁺	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
K ⁺	10.91	13.53	21.43	72.73	0.45	0.16	0.52	0.34
Na ⁺	79.23	98.13	155.4	525.9	3.64	1.32	4.37	2.87
Mg ⁺²	20.86	26.06	41.39	142.6	0.06	0.02	0.07	0.05
Ca ⁺²	62.15	77.63	123.3	424.7	0.23	0.08	0.28	0.18
Sr ⁺²	0.34	0.42	0.67	2.32	0.00	0.00	0.00	0.00
Ba ⁺²	0.08	0.10	0.16	0.55	0.00	0.00	0.00	0.00
CO ₃ ⁻²	0.62	0.96	2.36	22.35	0.00	0.00	0.00	0.00
HCO ₃ ⁻	167.8	207.8	327.4	1,080	5.86	2.22	7.13	4.71
NO ₃ ⁻	32.67	40.35	63.84	214.6	1.97	0.73	2.39	1.58
F ⁻	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Cl ⁻	170.0	212.0	336.5	1,156	2.03	0.70	2.35	1.54
Br ⁻¹	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SO ₄ ⁻²	21.00	26.23	41.65	143.5	0.08	0.03	0.09	0.06
PO ₄ ⁻³	0.95	1.18	1.88	6.46	0.01	0.00	0.01	0.01
SiO ₂	34.20	42.60	67.59	231.5	0.62	0.21	0.74	0.48
Boron	0.07	0.08	0.10	0.21	0.05	0.03	0.06	0.05
CO ₂	5.21	5.34	5.81	8.89	5.42	5.81	8.96	7.40
TDS*	601.0	747.3	1,184	4,027	14.99	5.51	18.01	11.87
Cond. μS/cm	941	1,160	1,798	5,643	21	8	25	16
pH	7.7	7.7	7.9	8.1	6.3	5.9	6.2	6.1

Footnotes:

*Total Dissolved Solids and Conductivity includes ions, SiO₂ and B. It does not include NH₃ and CO₂

RO Design Warnings

None

Special Comments

None

RO Flow Table (Element Level) - Pass 1

Stage	Element	Element Name	Recovery (%)	Feed Flow (m³/h)	Feed Press (bar)	Feed TDS (mg/L)	Conc Flow (m³/h)	Perm Flow (m³/h)	Perm Flux (LMH)	Perm TDS (mg/L)
PF	1	*FilmTec™ SOAR 5000iEE4	3.1	6.17	2.4	601.1	5.98	0.19	5.1	12.75
PF	2	*FilmTec™ SOAR 5000iEE4	3.1	5.98	2.4	619.8	5.80	0.19	5.0	13.46
PF	3	*FilmTec™ SOAR 5000iEE4	3.1	5.80	2.3	639.2	5.62	0.18	4.9	14.21
PF	4	*FilmTec™ SOAR 5000iEE4	3.1	5.62	2.3	659.3	5.44	0.18	4.7	15.00
PF	5	*FilmTec™ SOAR 5000iEE4	3.2	5.44	2.2	680.1	5.27	0.17	4.6	15.83
PF	6	*FilmTec™ SOAR 5000iEE4	3.2	5.27	2.2	701.7	5.10	0.17	4.5	16.70
PF	7	*FilmTec™ SOAR 5000iEE4	3.2	5.10	2.1	724.1	4.94	0.16	4.4	17.62
CC1	1	*FilmTec™ SOAR 5000iEE4	6.3	12.1	9.3	685.4	11.4	0.76	20.5	4.24
CC1	2	*FilmTec™ SOAR 5000iEE4	6.6	11.4	9.2	731.0	10.6	0.75	20.2	4.57
CC1	3	*FilmTec™ SOAR 5000iEE4	7.0	10.6	9.1	782.2	9.88	0.74	20.0	4.95
CC1	4	*FilmTec™ SOAR 5000iEE4	7.4	9.88	9.0	840.5	9.15	0.73	19.7	5.39
CC1	5	*FilmTec™ SOAR 5000iEE4	7.9	9.15	9.0	907.3	8.42	0.73	19.5	5.89
CC1	6	*FilmTec™ SOAR 5000iEE4	8.5	8.42	8.9	984.8	7.71	0.72	19.3	6.48
CC1	7	*FilmTec™ SOAR 5000iEE4	9.2	7.71	8.8	1,076	7.00	0.71	19.1	7.17
CC Final	1	*FilmTec™ SOAR 5000iEE4	6.5	12.1	10.7	2,427	11.4	0.78	21.1	13.16
CC Final	2	*FilmTec™ SOAR 5000iEE4	6.8	11.4	10.6	2,593	10.6	0.77	20.7	14.43
CC Final	3	*FilmTec™ SOAR 5000iEE4	7.1	10.6	10.5	2,780	9.83	0.75	20.2	15.88
CC Final	4	*FilmTec™ SOAR 5000iEE4	7.5	9.83	10.5	2,991	9.10	0.74	19.8	17.57
CC Final	5	*FilmTec™ SOAR 5000iEE4	7.9	9.10	10.4	3,231	8.38	0.72	19.4	19.56
CC Final	6	*FilmTec™ SOAR 5000iEE4	8.4	8.38	10.3	3,505	7.68	0.70	18.9	21.92
CC Final	7	*FilmTec™ SOAR 5000iEE4	8.9	7.68	10.2	3,822	7.00	0.68	18.4	24.78

Footnotes:

*Total Dissolved Solids and Conductivity includes ions, SiO₂ and B. It does not include NH₃ and CO₂

RO Solubility Warnings

Warning	Pass No
Langelier Saturation Index > 0	1
BaSO ₄ (% saturation) > 100	1
SiO ₂ (% saturation) > 100	1
Anti-scalants may be required. Consult your anti-scalant manufacturer for dosing and maximum allowable system recovery.	1

RO Chemical Adjustments

	Pass 1 Feed	RO 1 st Pass Conc
pH	7.7	8.1
Langelier Saturation Index	-0.03	2.00
Stiff & Davis Stability Index	0.56	1.88
TDS ^a (mg/l)	601.0	4,028
Ionic Strength (molal)	0.01	0.08
HCO ₃ ⁻ (mg/L)	167.8	1,081
CO ₂ (mg/l)	5.21	8.89
CO ₃ ⁻² (mg/L)	0.62	22.34
CaSO ₄ (% saturation)	0.30	6.6
BaSO ₄ (% saturation)	55.1	769.7
SrSO ₄ (% saturation)	0.17	1.7
CaF ₂ (% saturation)	0.00	0.00
SiO ₂ (% saturation)	29.7	176.3
Mg(OH) ₂ (% saturation)	0.00	0.09

Footnotes:

*Total Dissolved Solids and Conductivity includes ions, SiO₂ and B. It does not include NH₃ and CO₂

RO Utility and Chemical Costs

Service Water

	Flow Rate (m ³ /h)	Unit Cost (\$/m ³)	Hourly Cost (\$/h)	Daily Cost (\$/d)
Non-Product Feed Water				
Pass 1	92.9	0.1400	13.01	312.25
Total Non-product Feed Water Cost	92.9		13.01	312.25
Waste Water Disposal				
Pass 1	92.9	0.6900	64.12	1538.96
Total Waste Water Disposal	92.9		64.12	1538.96
Total Service Water Cost				1851.21

Electricity

Peak Power	(kW)	423.7
Energy	(kWh/d)	6,609
Electricity Unit Cost	(\$/kWh)	0.0900
Electricity Cost	(\$/d)	594.8
Specific Energy	(kWh/m ³)	0.51

Pump	Flow Rate (m³/h)	Pressure (bar)	Power (kW)	Energy (kWh/d)	Cost (\$/d)
Pass 1					
Boost (CC)	617.43	10.82 - 12.24	232.58 - 263.06	5,016.42	451.48
Boost (PF)	740.8	3.9	100.61	378.72	34.08
Recirculation	840.00	2.05	59.99	1,213.86	109.25
Pass 1 Total			423.66	6,609.00	594.81
System Total			423.66	6,609.00	594.81

Chemical

Chemical	Unit Cost (\$/kg)	Dose (mg/L)	Volume (L/d)	Cost (\$/d)
Total Chemical Cost				0.0

Utility and Chemical Cost	(\$/d)	2,446
Specific Water Cost	(\$/m ³)	0.187

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